

CASE STUDY 01 / HEALTHCARE

Excelss healthcare platform

Team leadership and independent delivery across a nurse/client management ecosystem.

Laravel

jQuery

NestJS

Angular 18

Twilio

CHALLENGE

Support operational healthcare workflows while evolving communication, integrations and distinct application components.

PERSONAL ROLE

Led and mentored a five-member team on the Laravel and jQuery platform. Separately owned and delivered a subsystem built with NestJS and Angular 18.

ENGINEERING ACTIONS

- Built application workflows and reusable Angular UI modules, connecting frontend experiences with backend APIs.
- Implemented real-time communication using Twilio and improved chat behavior under load.
- Integrated SurveyMonkey and Buzz Communications; refined API payloads and database indexing.

TWO DELIVERY STREAMS, SHARED PRODUCT CONTEXT

Team-led platform

Laravel + jQuery

Owned subsystem

NestJS + Angular 18

Service integrations

Twilio / SurveyMonkey / Buzz

Simplified architectural illustration - not a proprietary diagram or screenshot.

DELIVERED CAPABILITY

A connected nurse/client platform with operational workflows, real-time communication and external-service integrations, supported by a separately delivered modern application subsystem.

AVAILABLE FOR REMOTE ROLES & CONTRACT PROJECTS

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CASE STUDY 02 / ANALYTICAL SYSTEMS

HydroGeologic platform

Connecting an interactive web application to Python-driven data optimization workflows.

Node.js

Angular

Python engine

AWS SQS

WebSockets

Highcharts

CHALLENGE

Heavy background calculations needed a web interface that could remain responsive while modelling work progressed.

PERSONAL ROLE

Designed and delivered the Node.js and Angular application interface, asynchronous processing approach and live dashboard integration.

ENGINEERING ACTIONS

- Connected the Node.js and Angular platform to Python-driven data optimization engines.
- Used AWS SQS to move substantial modelling operations into asynchronous background processing.
- Combined WebSockets and Highcharts to deliver live analytical data to frontend dashboards.

ASYNCHRONOUS MODELLING WITH LIVE PRESENTATION



Live dashboard updates via WebSockets

Simplified architectural illustration - not a proprietary diagram or screenshot.

DELIVERED CAPABILITY

A web-based analytical experience that separates lengthy modelling work from synchronous requests and presents live data through interactive dashboards.

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CASE STUDY 03 / LEGALTECH

Legalshark application layers

A modular .NET/NOP architecture for maintainable, separated legal workflows.

.NET / NOP

Modular architecture

Cross-layer APIs

Plugins

CHALLENGE

Framework rigidity and tightly connected application layers created obstacles for data isolation, case management and ongoing feature work.

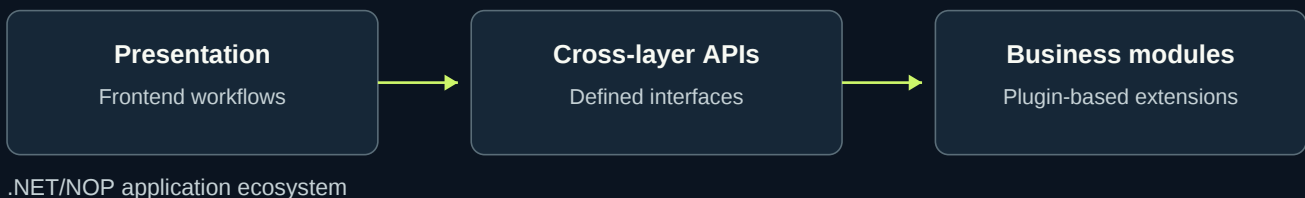
PERSONAL ROLE

Acted as lead full stack architect and developer, defining application boundaries and implementing the APIs and extension structure.

ENGINEERING ACTIONS

- Designed a decoupled application-layer framework within the .NET/NOP ecosystem.
- Created cross-layer APIs that separate business logic from presentation and support legal workflow data flows.
- Used modular, plugin-based architecture to work around framework limitations and enable feature evolution.

EXPLICIT BOUNDARIES AND EXTENSIBLE BUSINESS LOGIC



Simplified architectural illustration - not a proprietary diagram or screenshot.

DELIVERED CAPABILITY

A clearer separation of presentation and business logic, with defined API boundaries and a modular foundation for evolving case-management features.

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CASE STUDY 04 / APPLIED COMPUTER VISION

Fabric AI

Computer vision for fabric defect inspection

Independent project · In development

Python

Ultralytics YOLOv8

OpenCV

Gradio

ReportLab

CHALLENGE

Turn fabric images and recorded video into reviewable defect detections, while supporting repeatable local training and dataset preparation.

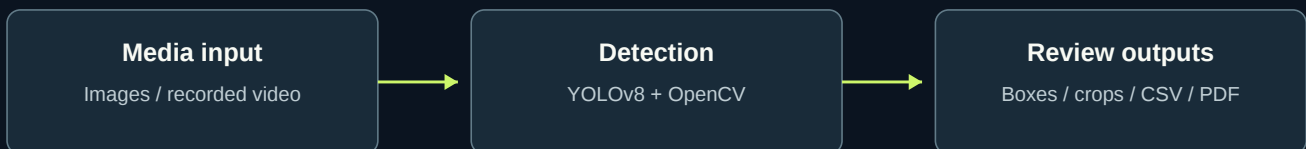
PERSONAL ROLE

Building the Python application across inference, training workflows, data utilities and an operator-facing Gradio interface. Local training and inference runs are recorded in the project.

ENGINEERING ACTIONS

- Implemented YOLOv8 inference with bounding boxes, confidence labels and saved defect crops for images and recorded video.
- Built Gradio tabs for Detect, Train, Dataset, Evaluate and Settings, with CSV exports and PDF inspection reports.
- Added checkpoint fine-tuning and resume, dataset validation, polygon-to-box conversion and per-class confidence controls.

FROM INPUT TO REVIEWABLE DETECTIONS



Gradio workspace: detection, training, dataset tools, evaluation and settings

Illustrative workflow - not a detection screenshot or measured model result.

CURRENT CAPABILITY

A working local inspection workflow connects custom-model inference, training controls and reviewable outputs. Development continues on model quality and workflow refinement.

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